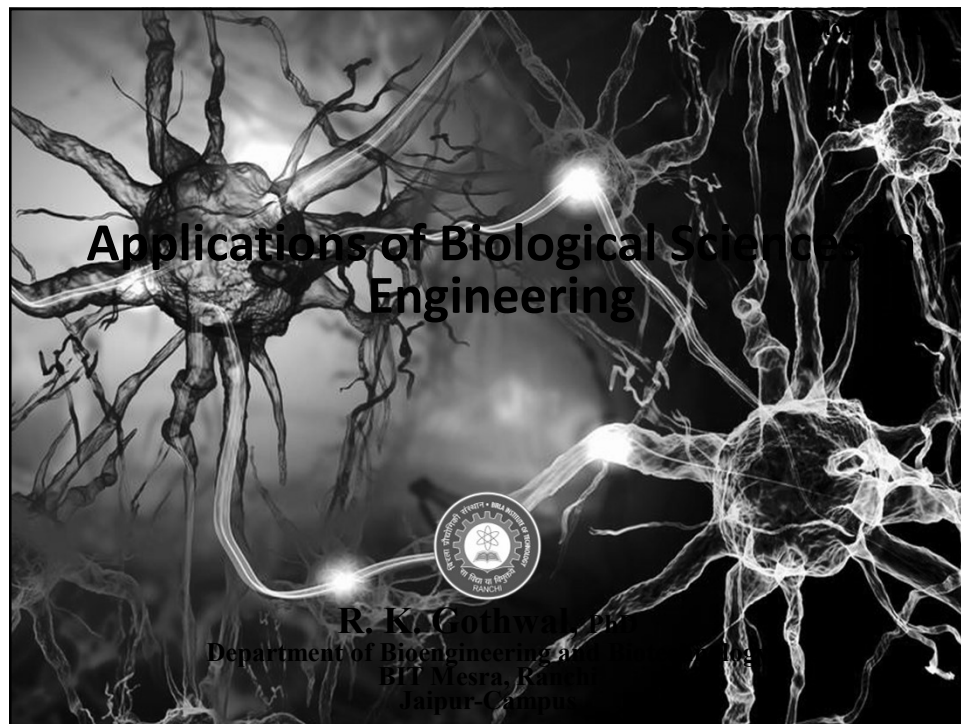




Applications of Biological Sciences in Engineering

- Biomaterials, Bioinformatics, Biosensors and Bioelectronics (Biological Sensors – Ear & Eye)
- Synthetic Biology
- Nanotechnology

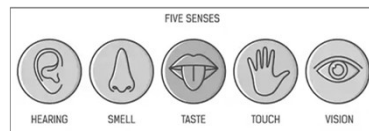
Biological signal

- **Biological Signal or Biosignal** is the signal produced by the living organisms which can be measured and analyzed by biomedical devices. For example, heart sound can be listened by stethoscope.
- Biosignals can be biomedically defined as “any **space-time record** of biological event which can be measured and analyzed”.
- These signals are mostly **electrical** or **electrochemical** in nature.
- Any biological event (including **electrical, chemical or mechanical activity**) generates signals of the respective events which can be measured and analyzed.
- These Biosignals, therefore, contain **useful information** that can be used to understand the underlying physiological mechanisms of a specific biological event or system and that may be useful for medical diagnosis.
- Biosignals are observations of physiological activities of organisms, ranging from gene and protein sequences, to neural and cardiac rhythms, to tissue and organ images.

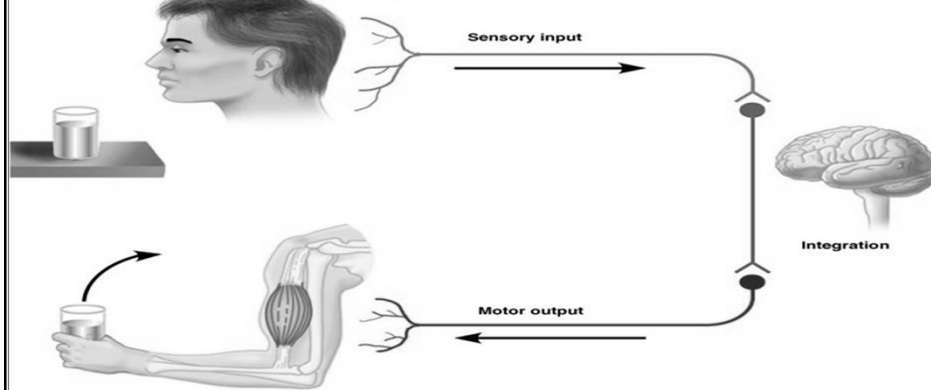


Brain: The human microcontroller

- Brain in human body controls various physiological and mechanical activities through nervous system.
- Humans have five basic **senses**: sight, hearing, smell, taste and touch.
- Living organisms make use of their senses to acquire information from outside environment and communicate this information to the brain to take action.
- The sensory organs transform these signals into electro-chemical message and convey these messages to brain in the form of nerve impulse (action potential).
 - **Sensory Nerve cell** (neuron) sends impulse to the brain.
 - **Motor neuron** brings back the triggered action from the brain (what to do).



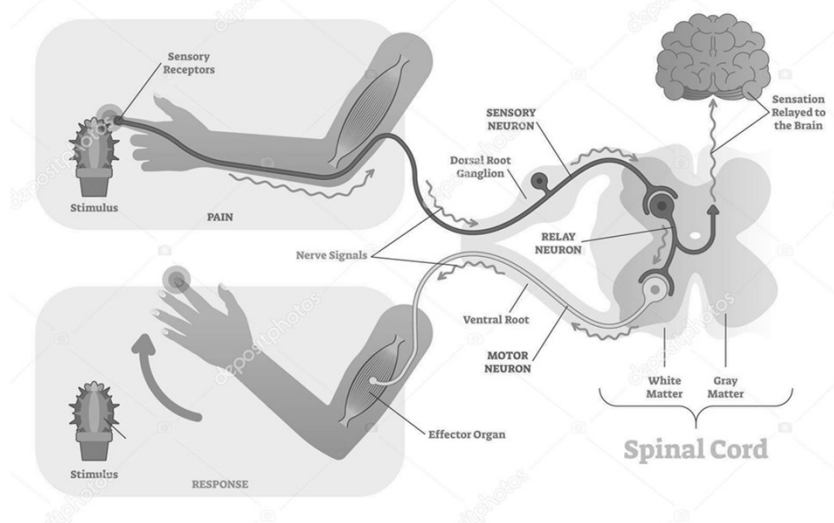
How an impulse is Transferred



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A stimulus occurs and the sensory neurons send electrical impulses to the brain. The brain signals and sends electrical impulses to the motor neurons which trigger a contraction in the arm to reach for the water.

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Biosignals: electrical / non-electrical

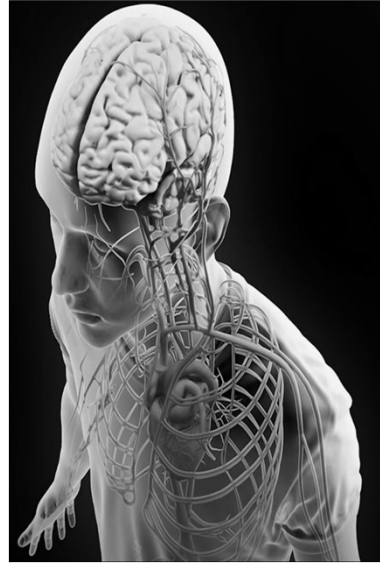
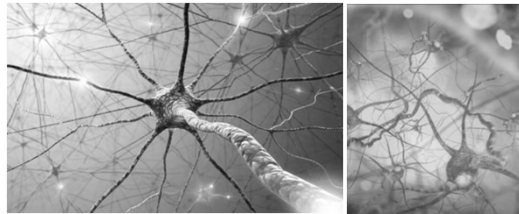
- The term **biosignal** is often used to refer to bioelectrical signals, but it may refer to both electrical and non-electrical signals.
- Non-electrical biosignals are capable of being monitored from biological beings, such as
 - mechanical signals (e.g. the mechanomyogram or MMG),
 - acoustic signals (e.g. phonetic and non-phonetic utterances, breathing),
 - chemical signals (e.g. pH, oxygenation, metabolites)
 - optical signals (e.g. movements).

Electrical biosignals

- Electrical biosignals, or bioelectrical time signals, usually refers to the **change in electric current produced by the sum of an electrical potential difference across a specialized tissue, organ or cell system** like the nervous system.
- Among the best-known bioelectrical signals are:
 - Electroencephalogram (EEG) – Signals in Brain
 - Electrocardiogram (ECG) – Signals in Heart
 - Electromyogram (EMG) – Signals in Muscles
 - Electrooculography (EOG) – Signals in Eyes
 - Galvanic skin response (GSR) – electrical signals in skin,
 - Mechanomyogram (MMG) – mechanical signal of muscles
 - Magnetoencephalogram (MEG) – magnetic fields produced by electrical currents in brain
- EEG, ECG, EOG and EMG are measured with a differential amplifier which registers the difference between two electrodes attached to the skin.
- However, the galvanic skin response measures electrical resistance and the MEG measures the magnetic field induced by electrical currents (electroencephalogram) of the brain.

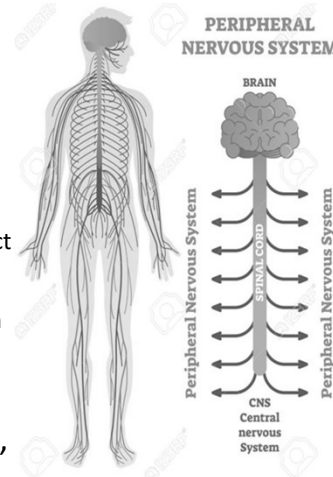
Nervous System

- The nervous system is an extraordinarily complex communication system that can send and receive voluminous amounts of information simultaneously.
- It is a complex collection of nerves and specialized cells known as **neurons** that transmit signals between different parts of the body.
- It is essentially the body's electrical wiring.



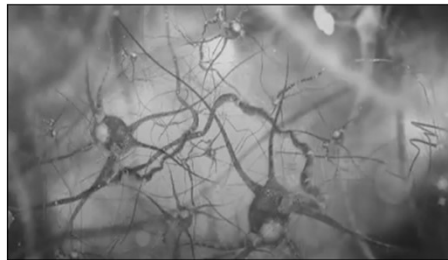
Nervous System

- Structurally, the nervous system has two components:
 - **Central nervous system (CNS)** and
 - It is made up of the **brain, spinal cord and nerves**.
 - **Peripheral nervous system (PNS)**
 - It consists of **sensory neurons, ganglia** (clusters of neurons) and **nerves** that connect to one another and to the central nervous system.
 - PNS includes all the nerves outside the brain and spinal cord such as arms, hands, legs, feet etc.
- **Autonomous Nervous System** – It connects CNS to Heart, Lung, Stomach, Bladder and Sex Organs.

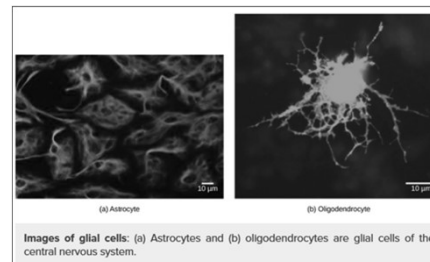


Cells of Nervous System

- **Neuron** (Nerve cell)
 - Structural and functional unit of nervous system. Communicate with other cells via synapses, that allows rapid transmission of signals, either electrical or chemical.
- **Glia** (Glial cells)
 - Known as “supporting cells” of nervous system as it provide support and protection for neurons



Neurons

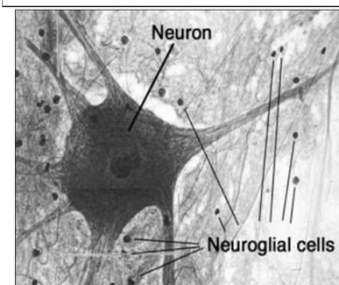
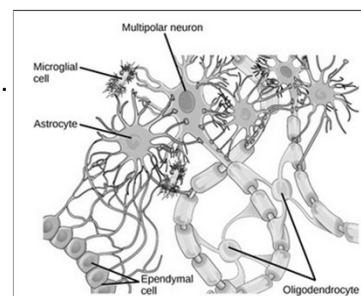


Images of glial cells: (a) Astrocytes and (b) oligodendrocytes are glial cells of the central nervous system.

Glial cells

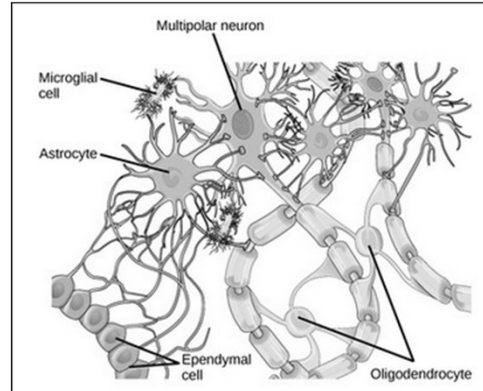
Glia (Glial cells)

- **Glia (Glial cells)**
 - derived from the Greek word for "glue".
 - Glial cells are **non-conducting cells** whose function is to **support** the nervous system and to protect or nourish neurons.
 - the number of glial cells in the brain actually outnumbers the number of neurons by a **factor of ten**.
 - Neurons would be unable to function without the vital roles that are fulfilled by these glial cells.
 - Glia guide developing neurons to their destinations, buffer ions and chemicals that would otherwise harm neurons, and provide myelin sheaths around axons.



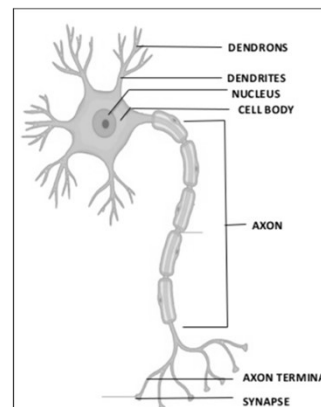
Glia (Glial cells)

- The four main functions of glial cells are:
 - to surround neurons and hold them in place,
 - to supply nutrients and oxygen to neurons,
 - to insulate one neuron from another, and
 - to destroy and remove the carcasses of dead neurons (clean up).



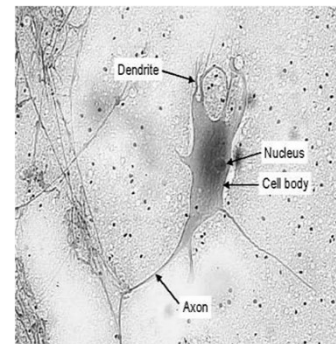
Neurons

- They are the structural and functional unit of the nervous system.
- They receive the sensory input from the external world and convey it to the brain and brings back the command information from brain back to effector to take action or response.
- **Neurons are the longest cell in our body.**
- Once fully developed, they never divide and remain in interface throughout life.
- They possess the unique property of excitability and conductivity together.
 - Excitability is the ability to initiate a nerve impulse in response to stimuli.
 - Conductivity is the ability to transmit a nerve impulse.
- It is roughly estimated that around **86 billion** neurons are present in our body, which comprises roughly 10% of all brain cells.



Neurons

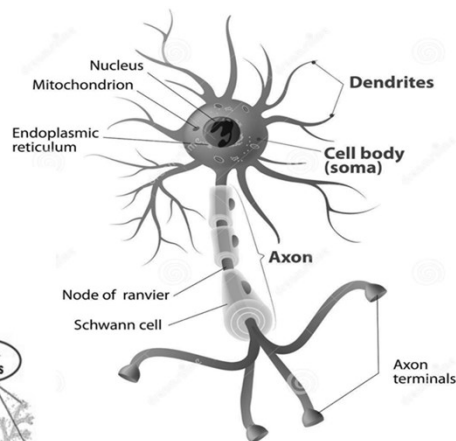
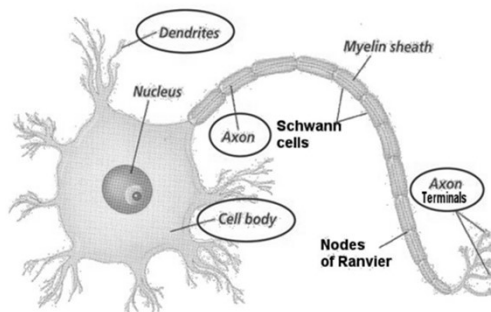
- Neuron is a microscopic structure composed of **Cell body**, **Dendrites** and **Axons**.
- Neurons are a specialized type of cells that integrates electrochemical activity of the other neurons that are connected to it and that propagates that integrated activity to other neurons.
- Most neurons share the same cellular components.
- But neurons are also highly specialized: different types of neurons have different sizes and shapes that relate to their functional roles.



Microscopic image of a neuron

Structure of Neurons (Nerve Cell)

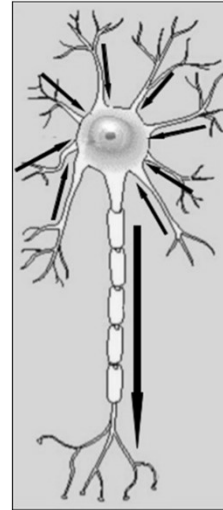
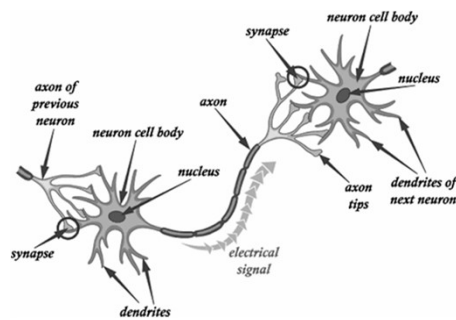
- Cell body (Soma)
- Dendrites
- Axon
- Axon Terminal



Structure of Neurons

Dendrites (single dendron)

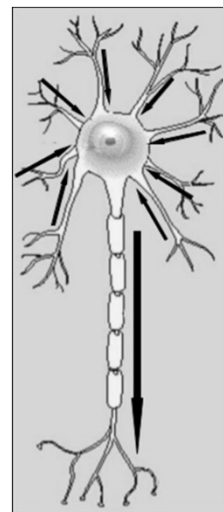
- The **Dendrites** (Greek, dendro: tree) of a **neuron** are its many short, branching fibers extending from the cell body or soma. They contain both neurofibrils and Nissl's granules. The branches are studded with spiny projections. They form the afferent component of impulse means carry impulse towards cell body.
- The function of a dendrite is to receive signals from other neurons, to process these signals, and to transfer the information to the soma of the neuron.



Structure of Neurons

Axon

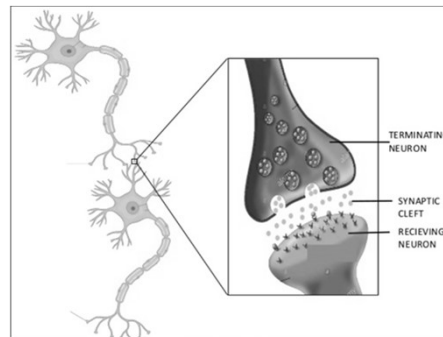
- An **axon** (from Greek, axis) is a long, slender projection of a neuron which conducts electrical impulses (action potentials) away from the nerve cell body to another neurons or muscles or glands. It forms the efferent component of impulse means carry impulse away from cell body. It contains neurofibrils and no Nissl's granules.
- Some types of neurons have no axon and transmit signals from their dendrites. The swollen end of an axon is known as the **axon terminal** which joins the dendron or cell body of another neuron forming a synaptic connection.
- An Axon Terminal is the end of a branch of a nerve's axon, conducts electrical signals to a nerve synapse.
- Axons make contact with other cells—usually other neurons but sometimes muscle or gland cells—at junctions called **synapses**.



Structure of Neurons

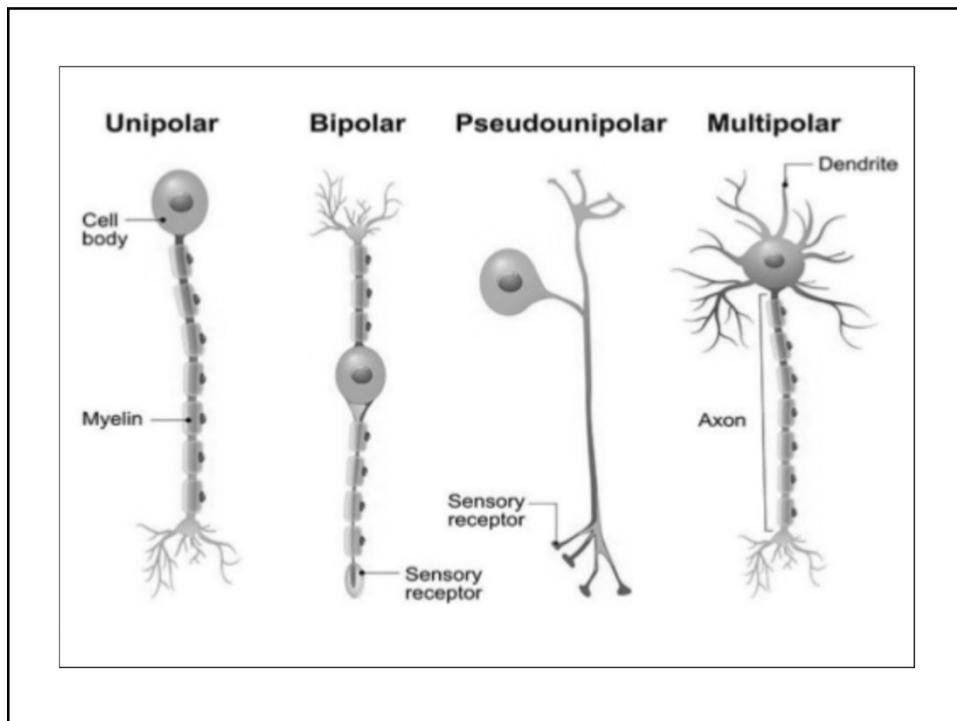
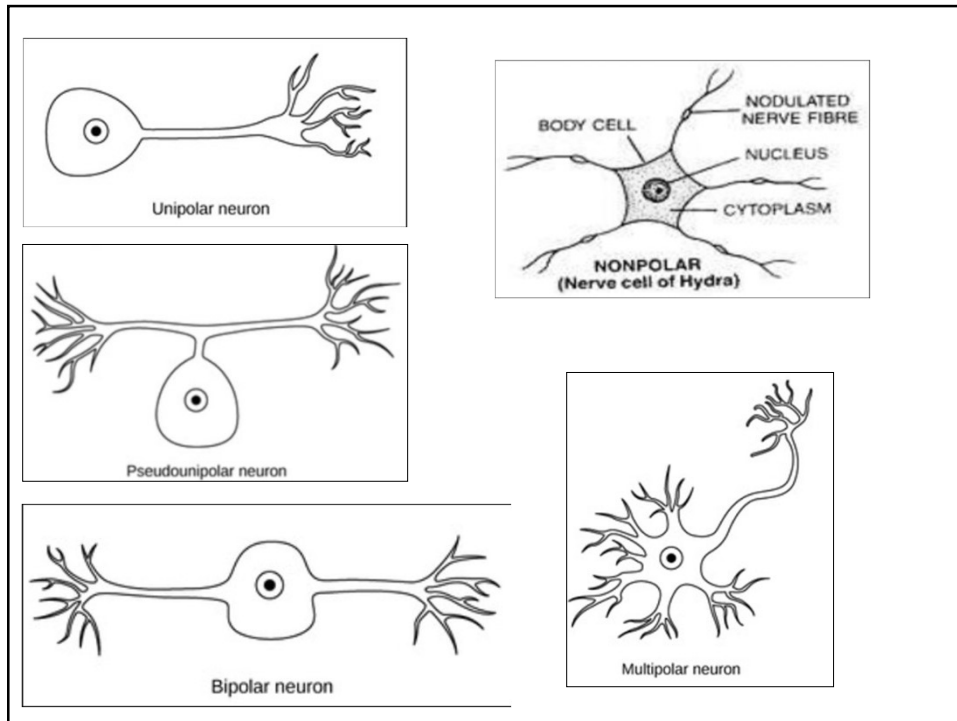
Synapse

- A synapse is the site of junction between an axon and the dendrite of another neuron.
- However they do not touch each other and remain separate by a microscopic gap of about 200 Angstrom.
- Each neuron receives an impulse through its dendrites and passes it to the next neuron through synapse. The impulse transmit through chemicals called **neurotransmitters**.
- **Acetylcholine** is the first neurotransmitter to be discovered.



Types of Neurons

- On the basis of **structure**, neurons may be classified into :
 - Nonpolar Neurons
 - There is no functional difference between axon and dendrons. Each processes can bring impulse to the cyton or can take away from cyton. These neurons are common in hydra but rare in vertebrates.
 - Unipolar Neurons
 - Unipolar neurons have only one structure that extends away from the cyton. These neurons are not found in vertebrates, but are found in insects where they stimulate muscles or glands.
 - Pseudounipolar Neurons
 - Pseudounipolar cells share characteristics with both unipolar and bipolar cells. A pseudounipolar cell has a single structure that extends from the soma (like a unipolar cell), which later branches into two distinct structures (like a bipolar cell).
 - Most sensory neurons are pseudounipolar and have an axon that branches into two extensions: one connected to dendrites that receives sensory information and another that transmits this information to the spinal cord.
 - Bipolar Neurons
 - A bipolar neuron has one axon and one dendrite extending from the soma. An example of a bipolar neuron is a retinal bipolar cell, which receives signals from photoreceptor cells that are sensitive to light and transmits these signals to ganglion cells that carry the signal to the brain.
 - Multipolar Neurons
 - Multipolar neurons are the most common type of neuron. Each multipolar neuron contains one axon and multiple dendrites. Multipolar neurons can be found in the central nervous system (brain and spinal cord).



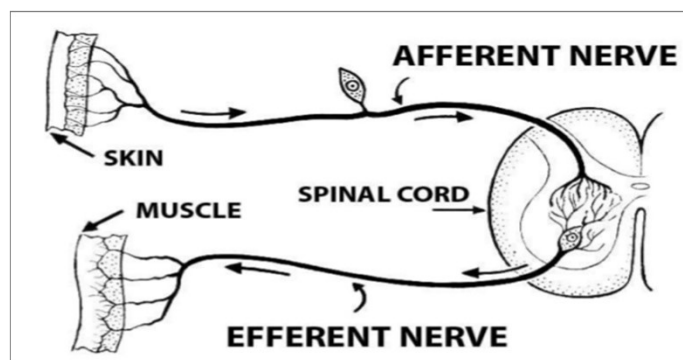
Types of Neurons

- On the basis of **function**, neurons may be classified into :
 - Sensory Neurons
 - Motor Neurons
 - Interneurons
- **Sensory neurons**
 - These are receptor neurons which connects sense organs with the CNS. It brings information from our sensory organs (senses) to Brain.
 - It also includes, all of the *cognitive* information through which we are able to reason, to think, to dream, to plan, to remember, and to do everything else that we do with our minds.
- **Motor neurons**
 - It carry information from Brain through which we are able to move and respond.
- **Interneurons**
 - These are connector neurons which are present in the CNS and occur between sensory and motor neurons for distance transmission of impulse.

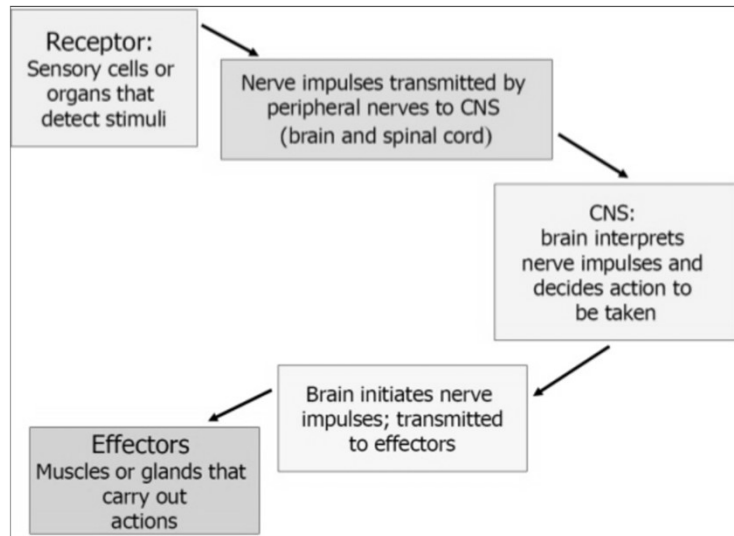
Types of Neurons

On the basis of **pathway of impulse**, neurons may be classified into :

- **Afferent Neurons (Sensory)**
 - These are sensory neurons that carry impulses from sense organ to CNS
- **Efferent Neurons(Motor)**
 - These are motor neurons which carry impulses from CNS to effector organ



Propagation of Nerve Impulse



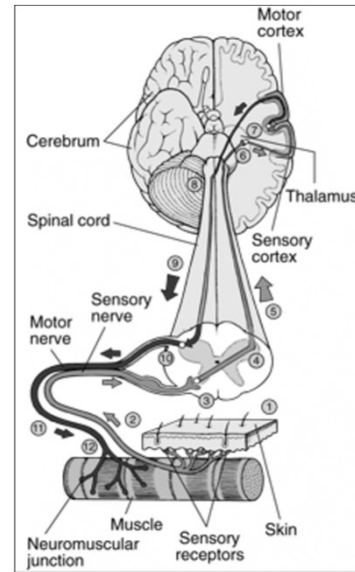
How Brain Move a Muscle

Example of Signal Propagation

- Moving a muscle usually involves communication between the muscle and the brain through nerves. The impetus to move a muscle may originate with the senses.
- For example, special nerve endings in the skin (sensory receptors) enable people to sense pain, when they step on a sharp rock, or to sense temperature, when they pick up a very hot cup of coffee. This information is sent to the brain, and the brain send a message to the muscle about how to respond.
- This type of exchange involves two complex nerve pathways:
 - The sensory nerve pathway to the brain
 - The motor nerve pathway to the muscle

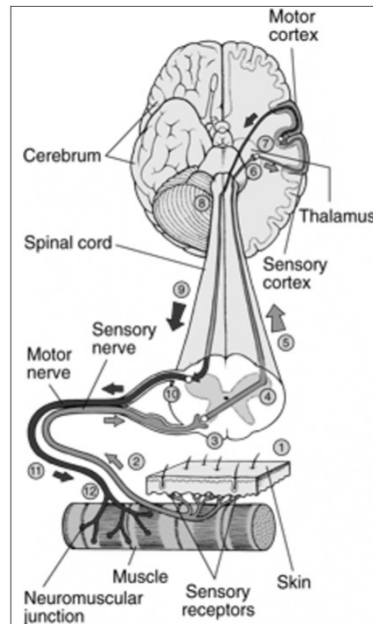
How Brain Move a Muscle

1. If sensory receptors in the skin detect pain or a change in temperature, they transmit an impulse (signal), which ultimately reaches the brain.
2. The impulse travels along a sensory nerve to the spinal cord.
3. The impulse crosses a synapse (the junction between two nerve cells) between the sensory nerve and a nerve cell in the spinal cord.
4. The impulse crosses from the nerve cell in the spinal cord to the opposite side of the spinal cord.
5. The impulse is sent up the spinal cord and through the brain stem to the **thalamus**, which is a **processing center** for sensory information, located deep in the brain.



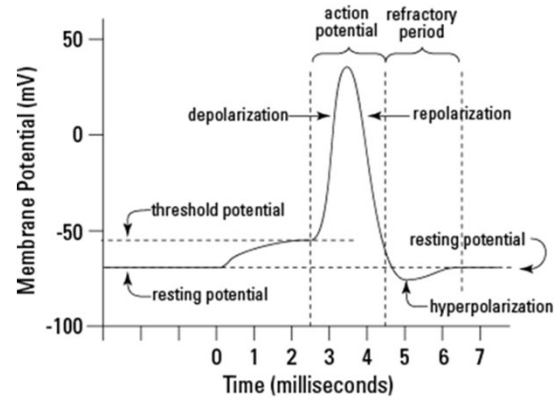
How Brain Move a Muscle

6. The impulse crosses a synapse in the thalamus to nerve fibers that carry the impulse to the sensory cortex of the cerebrum (the area that receives and interprets information from sensory receptors).
7. The sensory cortex perceives the impulse. A person may then decide to initiate movement, which triggers the motor cortex (the area that plans, controls, and executes voluntary movements) to generate an impulse.
8. The nerve carrying the impulse crosses to the opposite side at the base of the brain.
9. The impulse is sent down the spinal cord.
10. The impulse crosses a synapse between the nerve fibers in the spinal cord and a motor nerve, which is located in the spinal cord.
11. The impulse travels out of the spinal cord along the length of the motor nerve.
12. At the neuromuscular junction (where nerves connect to muscles), the impulse crosses from the motor nerve to receptors on the motor end plate of the muscle, where the impulse stimulates the muscle to move.



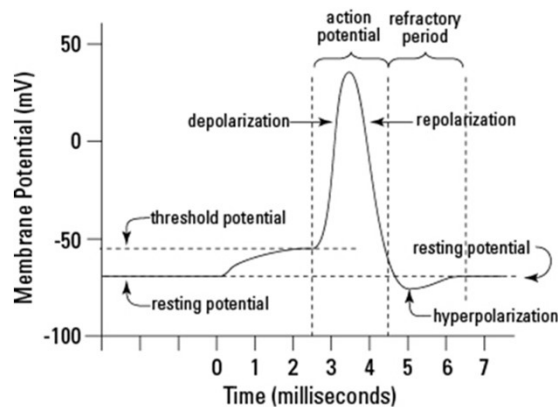
Mechanism of nerve impulse Propagation

- **Resting potential.** The resting potential describes the unstimulated, polarized state of a neuron (at about -70 millivolts).



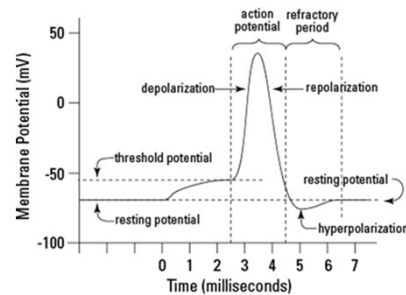
Mechanism of nerve impulse Propagation

- **Graded potential.** It is a change in the resting potential of the plasma membrane in the response to a stimulus.

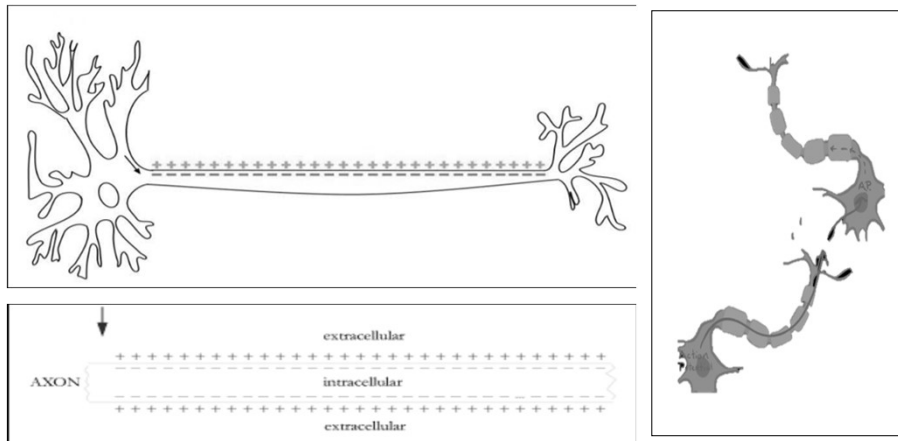


Graded potential

- A graded potential is a change in the resting potential of the plasma membrane in the response to a stimulus.
 - A graded potential occurs when the stimulus causes Na^+ or K^+ gated channels to open. If Na^+ channels open, positive sodium ions enter, and the membrane depolarizes (becomes more positive).
 - If the stimulus opens K^+ channels, then positive potassium ions exit across the membrane and the membrane **hyperpolarizes** (becomes more negative).
 - A graded potential is a local event that does not travel far from its origin.
 - Graded potentials occur in cell bodies and dendrites.
 - Light, heat, mechanical pressure, and chemicals, such as neurotransmitters, are examples of stimuli that may generate a graded potential (depending upon the neuron).



Mechanism of Nerve Impulse Propagation



Transmission of a nervous impulse across a synapse

1. Firstly, an action potential (change in electrical potential) arrives at the pre-synaptic neurone.
2. This changes the voltage in the neurone causing the voltage-gated calcium channels on the pre-synaptic neurone to open.
3. Calcium ions then diffuse into the pre-synaptic neurone.
4. The increased concentration of Calcium in the neurone then causes synaptic vesicles, containing the neurotransmitter acetylcholine, to move towards the membrane on the pre-synaptic neurone.
5. The vesicles fuse to the membrane and the neurotransmitter is released into the gap between the two neurones (known as the synaptic cleft).

Transmission of a nervous impulse across a synapse

6. The acetylcholine neurotransmitter then diffuses across the synaptic cleft towards the post-synaptic neurones membrane.
7. Here, the acetylcholine neurotransmitter then binds to the complimentary receptors on the post-synaptic neurone's membrane.
8. The increase in concentration of the neurotransmitter causes ligand (chemical) gated sodium channels in the post-synaptic neurone membrane to open, allowing sodium to diffuse into the post-synaptic neurone.
9. The increased concentration of sodium ions now in the post-synaptic neurone depolarise the neurone's membrane causing EPSPs (excitatory post-synaptic potentials).
10. If these EPSPs reach a certain threshold, then an action potential is initiated in the post-synaptic neurone and the impulse has been successfully transmitted from one neurone to the next.

Signal termination

- Termination of the signal lets the postsynaptic cell return to its normal resting potential, ready for new signals to arrive.
- **For the signal to end, the synaptic cleft must be cleared of neurotransmitter.**
- There are a few different ways to get this done.
 - The neurotransmitter may be broken down by an enzyme, it may be sucked back up into the presynaptic neuron, or it may simply diffuse away.
 - In some cases, neurotransmitter can also be "mopped up" by nearby glial cells.

Thanks